

CS 105: Department Introductory Course on Discrete Structures

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Lecture 15 – Counting and Combinatorics

Counting and Combinatorics

Topics to be covered

- ▶ Basics of counting
 - ▶ Product principle
 - ▶ Sum principle
 - ▶ Bijection principle
 - ▶ Double counting
- ▶ Subsets, partitions, Permutations and combinations
 1. Binomial coefficients and Binomial theorem
 2. Pascal's triangle
 3. Permutations and combinations with repetitions
 4. Estimating $n!$
- ▶ Today: Recurrence relations and generating functions

Solving general linear recurrence sequences

- ▶ (C.W) Find a solution for $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2, a_1 = 7$? Is $a_{25} > 2^{25}$?
- ▶ Find a solution for $a_n = 6a_{n-1} - 9a_{n-2}$, with $a_0 = 1, a_1 = 6$? Can you apply the same method for this? What went wrong?
- ▶ Recall the recurrence for Catalan Numbers: $C(n) = \sum_{i=1}^{n-1} C(i)C(n-i)$ for $n > 1, C(0) = 0, C(1) = 1$.
No. of ways to bracket a sum of n terms s.t. it can be computed by adding two numbers at a time?

This method does not work if we have repeated roots (this can be fixed!) and non-linear recurrences.

Reading assignment

Read examples/generalizations from Sections 6.1 and 6.2 from Rosen's book (7th Edition).

Proof Method 2: Using generating functions

Fibonacci recurrence relation

For $n \geq 2$, $F_n = F_{n-1} + F_{n-2}$, $F_0 = F_1 = 1$.

Compute F_n in terms of n . Recall: $F_n = 1, 1, 2, 3, 5, \dots$

► Consider the power series... $\phi(t) = \sum_{n=0}^{\infty} F(n)t^n$.

That is, $\phi(t) = 1 \cdot t^0 + 1 \cdot t^1 + 2 \cdot t^2 + 3 \cdot t^3 + 5 \cdot t^4 \dots$

$$t\phi(t) = \sum_{n=0}^{\infty} F(n)t^{n+1} = \sum_{n=1}^{\infty} F(n-1)t^n$$

$$t^2\phi(t) = \sum_{n=0}^{\infty} F(n)t^{n+2} = \sum_{n=2}^{\infty} F(n-2)t^n$$

$$(t + t^2)\phi(t) = t + \sum_{n=2}^{\infty} (F(n-1) + F(n-2))t^n = \sum_{n=0}^{\infty} F(n)t^n - 1$$

$$(t + t^2)\phi(t) = \phi(t) - 1$$

Proof Method 2: Using generating functions

- ▶ $\phi(t) = \frac{1}{1-t-t^2} = \frac{1}{(1-\alpha t)(1-\beta t)} = \frac{a}{1-\alpha t} + \frac{b}{1-\beta t}$
- ▶ Solving we get $\alpha = \frac{1+\sqrt{5}}{2}, \beta = \frac{1-\sqrt{5}}{2}, a = \frac{\sqrt{5}+1}{2\sqrt{5}}, b = \frac{\sqrt{5}-1}{2\sqrt{5}}$
- ▶ Now $\phi(t) = a(1 + \alpha t + \alpha^2 t^2 + \dots) + b(1 + \beta t + \beta^2 t^2 + \dots)$
- ▶ Equating coefficients of t^n we get $F_n = a\alpha^n + b\beta^n$.

thus, as before

$$F(n) = \frac{\sqrt{5}+1}{2\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n + \frac{\sqrt{5}-1}{2\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

Properties of generating functions

Definition

The (ordinary) generating function for a sequence $a_0, a_1, \dots \in \mathbb{R}$ is the infinite series $\phi(x) = \sum_{k=0}^{\infty} a_k x^k$.

- ▶ Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$, $g(x) = \sum_{k=0}^{\infty} b_k x^k$. Then
 1. If $f(x) = g(x)$, then $a_k = b_k$ for all k .
 2. $f(x) + g(x) = \sum_{k=0}^{\infty} (a_k + b_k) x^k$,
 3. $f(x)g(x) = \sum_{k=0}^{\infty} (\sum_{j=0}^k a_j b_{k-j}) x^k$,
 4. $\frac{d}{dx} (\sum_{k=0}^{\infty} a_k x^k) = \sum_{k=1}^{\infty} (k a_k) x^{k-1}$
- ▶ Let $u \in \mathbb{R}$, $k \in \mathbb{Z}^{\geq 0}$, Then extended binomial coefficient $\binom{u}{k}$ is defined as
$$\binom{u}{k} = \frac{u(u-1)\dots(u-k+1)}{k!} \text{ if } k > 0 \text{ and } = 1 \text{ if } k = 0.$$
- ▶ What if $u = -n$ for $n \in \mathbb{N}$?

The extended binomial theorem

Let $u \in \mathbb{R}$, $(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$.

If you don't like this, take $x \in \mathbb{R}, |x| < 1$.

Simple examples using generating functions

Standard identities:

- ▶ $\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k$
- ▶ $\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk}$
- ▶ $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

Class work/Pop Quiz!:

1. Solve the recurrence $a_k = 4a_{k-1}$ with $a_0 = 3$.
2. **A coding problem:** A cryptographer builds a system where a string of decimals is codeword if it contains an even number of 0s. E.g., 1023038 is valid but not 10244.

Let a_n be the number of n -digit codewords. Empty string is also valid.

- 2.1 Find a recurrence relation for a_n . What is a_0, a_1 ?
- 2.2 Solve the recurrence using generating functions.

Some applications of generating functions

- ▶ What is the number of ways a_k of selecting k elements from an n element set if repetitions are allowed?
 - ▶ Let $\phi(x) = \sum_{k=0}^{\infty} a_k x^k$.
 - ▶ Observe that $\phi(x) = (1 + x + x^2 + \dots)^n = (1 - x)^{-n}$
 - ▶ Expand this by the extended binomial theorem and compare coefficients of x^k .
 - ▶ $a_k = \binom{-n}{k}(-1)^k = (-1)^k \binom{n+k-1}{k}(-1)^k = \binom{n+k-1}{k}$.
 - ▶ (H.W) What if there must be ≥ 1 element of each type?
- ▶ Proving binomial identities: Show that $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$.
 - ▶ Compare coefficients of x^n in $(1+x)^{2n} = ((1+x)^n)^2$.
- ▶ (H.W) How many ways can a convex n -sided polygon be cut into triangles by adding non-intersecting diagonals (i.e., connecting vertices with non-crossing lines)? Write a recurrence and solve it!

Solving Catalan numbers using generating functions

Catalan Numbers

$$C(n) = \sum_{i=1}^{n-1} C(i)C(n-i) \text{ for } n > 1, C(0) = 0, C(1) = 1.$$

- ▶ Let $\phi(x) = \sum_{k=1}^{\infty} C(k)x^k$.
- ▶ Now consider $\phi(x)^2$.
- ▶
$$\begin{aligned}\phi(x)^2 &= (\sum_{k=1}^{\infty} C(k)x^k)(\sum_{k=1}^{\infty} C(k)x^k) \\ &= (\sum_{k=2}^{\infty} \sum_{i=1}^{k-1} C(i)C(k-i)x^k) \\ &= (\sum_{k=2}^{\infty} C(k)x^k) = \phi(x) - x\end{aligned}$$
- ▶ Solving for $\phi(x)$ we get, $\phi(x) = \frac{1}{2}(1 \pm (1 - 4x)^{1/2})$
- ▶ But since $\phi(0) = 0$, we have $\phi(x) = \frac{1}{2}(1 - (1 - 4x)^{1/2}) = \frac{1}{2} + (-\frac{1}{2}(1 - 4x)^{1/2})$.

Catalan numbers

Recall: Extended binomial theorem

Let $\alpha \in \mathbb{R}$, $(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$, where $\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}$.

- ▶ $\sum_{k=1}^{\infty} C(k)x^k = \phi(x) = \frac{1}{2} + (-\frac{1}{2}(1-4x)^{1/2}) = \frac{1}{2} + (-\frac{1}{2} \sum_{k=0}^{\infty} \binom{1/2}{k} (-4x)^k)$.
- ▶ The coefficient of x^k is $C(k) = -\frac{1}{2} \binom{1/2}{k} (-4)^k$
 $= -\frac{1}{2} (\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)\dots(\frac{1}{2}-k+1)) \frac{(-4)^k}{k!}$
 $= -\frac{1}{2} (\frac{1}{2})(-\frac{3}{2})(-\frac{5}{2})\dots(-\frac{2k-3}{2}) \frac{(-4)^k}{k!}$
- ▶ $C(k) = \frac{(-1)^k (-4)^k}{2^{k+1} k!} \cdot 1 \cdot 3 \dots (2k-3)$
- ▶ $C(k) = \frac{1 \cdot 4^k}{2^{k+1} \cdot k!} \cdot \frac{1 \cdot 2 \dots (2k-3)(2k-2)}{2^{k-1} (k-1)!} = \frac{(2k-2)!}{k!(k-1)!}$.

Thus, the n^{th} Catalan number is given by

$$C(n) = \frac{(2n-2)!}{n!(n-1)!} = \frac{1}{n} \binom{2n-2}{n-1}$$